

MASK WHAT MATTERS: CONTROLLABLE TEXT-GUIDED MASKING FOR SELF-SUPERVISED MEDICAL IMAGE ANALYSIS

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ABSTRACT

The scarcity of expert annotation remains a fundamental constraint on medical image representation learning, because labels are expensive, specialized, and often available only at limited scale. Self-supervised masked image modeling (MIM) offers a compelling response to this constraint, yet most existing methods still rely on random, high-ratio masking that is inefficient and only indirectly related to clinically meaningful image content. Region-aware variants partially mitigate this limitation, but they typically depend on reconstruction heuristics or supervised localization signals, which restricts their adaptability across tasks and imaging modalities. We introduce Mask What Matters, a controllable text-guided masking framework for externally guided self-supervised medical image pretraining. MWM uses a frozen biomedical vision-language model to localize prompt-specified regions and applies differentiated masking that concentrates reconstruction pressure on clinically relevant anatomy or pathology while reducing redundant masking of background tissue. This formulation makes semantic guidance an explicit control signal for MIM while avoiding per-image annotations during pretraining. Across brain MRI, chest CT, and lung X-ray experiments, Mask What Matters consistently outperforms existing MIM baselines, including SparK, with gains of up to +3.1 percentage points in classification accuracy, +1.3 in box average precision (BoxAP), and +1.1 in mask average precision (MaskAP) for detection. The same trend appears at substantially lower overall masking ratios, including 40% rather than the 70% setting favored by SparK. These results support controllable text-driven masking as an effective mechanism for semantically aligned medical image pretraining under limited annotation, while making explicit the method’s dependence on prompt-level semantic guidance and pretrained localization components.

1 INTRODUCTION

Image classification, segmentation, and object detection are central objectives in computer vision, and recent advances in these tasks have been driven largely by deep learning. The effectiveness of modern deep models, however, remains closely tied to the availability of large annotated datasets. In medical image analysis, where annotation requires domain expertise, careful quality control, and substantial clinical effort, this dependence remains a persistent barrier to robust and transferable visual representation learning.

Self-supervised learning (SSL) reduces this reliance on manual labels by pretraining models on unlabeled data and transferring the resulting representations to downstream tasks [Jaiswal et al. \(2020\)](#); [Liu et al. \(2021\)](#); [Krishnan et al. \(2022\)](#); [Huang et al. \(2023\)](#). Within SSL, MIM has become a particularly influential paradigm: the model observes a partially masked image and reconstructs the missing content from visible context. This objective encourages the encoder to capture structural and semantic regularities, and it has achieved strong downstream performance in both general vision and medical image analysis [He et al. \(2022\)](#); [Gupta et al. \(2025\)](#); [Xiao et al. \(2023\)](#).

Prevailing MIM methods, including masked autoencoders (MAE) [He et al. \(2022\)](#) and SparK [Tian et al. \(2023\)](#), nevertheless typically mask image regions at random. As a result, the reconstruction

signal is only weakly coupled to the regions that drive downstream decisions. This task-agnostic design has two consequences. First, representations may become semantically misaligned with the target objective, limiting transfer performance. Second, many methods adopt very high masking ratios (e.g., 75% in MAE) to increase the probability that informative regions are hidden, which raises reconstruction difficulty and places greater demands on data scale and computation. In contrast, masked language modeling benefits from the compact and semantically dense structure of text, enabling effective learning at lower masking ratios [Devlin et al., 2019]. The reliance of image-based MIM on high masking ratios therefore reflects both visual redundancy and the absence of semantic control over which regions are masked.

This limitation is especially consequential in medical imaging. Medical images are often semantically sparse: diagnostic evidence is concentrated in localized structures such as lesions, organs, or abnormal tissue, whereas large image regions may contain background or repetitive anatomy. At the same time, MRI, CT, and X-ray images differ substantially in appearance, spatial organization, and the scale of task-relevant regions. Effective medical MIM therefore requires masking strategies that are not merely random or high-ratio, but efficient, semantically targeted, and adaptable across the modalities and tasks for which such evidence is spatially concentrated.

We address these challenges with a controllable, text-guided masking framework, **Mask-What-Matters (MWM)**. Given a user-specified open-vocabulary phrase or description, a frozen biomedical vision-language model supplies zero-shot localization cues that MWM converts into regions of interest (ROIs). The framework then applies prompt-conditioned, region-specific masking: semantically important areas such as lesions or organs receive higher masking ratios, whereas background regions receive lower ratios. MWM thus incorporates prompt-level semantic intent into the pretraining signal without requiring per-image reports or dense annotations, and should be interpreted as externally guided self-supervised pretraining rather than as purely unguided SSL.

The paper makes three contributions that connect the proposed masking mechanism to the empirical evidence:

- A controllable text-guided MIM framework for medical imaging. MWM integrates vision-language localization (BiomedCLIP) with segmentation refinement (SAM) to derive prompt-relevant regions from open-vocabulary text and to apply differentiated ROI vs. background masking, directly targeting the semantic misalignment induced by random masking.
- A pretraining protocol that avoids per-image annotations while remaining compatible with multiple backbone families. MWM uses prompt-level semantic guidance rather than image-specific labels or reports, and the same masking interface can be incorporated into both ViT- and ConvNet-based MIM pipelines.
- Empirical evidence across modalities and downstream tasks. On brain MRI, chest CT, and X-ray datasets, MWM improves classification over strong MIM baselines such as SparK and improves detection and segmentation metrics, demonstrating the value of text-driven generative pretraining within the evaluated medical imaging settings.

2 RELATED WORK

2.1 MASKED IMAGE MODELING

Masked modeling was first established in natural language processing through the bidirectional encoder representations from transformers (BERT) model [Devlin et al., 2019], which learns contextual representations by masking tokens and predicting them during pretraining. Extending the same principle to vision, [He et al., 2022] introduced MIM, in which image regions are hidden and reconstructed from the remaining visible content. This paradigm has demonstrated strong transfer in natural and medical image domains [Gupta et al., 2025; Liu et al., 2021; Huang et al., 2023]. Representative methods include MAE [He et al., 2022], MedMAE [Gupta et al., 2025; Xiao et al., 2023], and SparK [Tian et al., 2023]: MAE uses high-ratio random masking to encourage ViTs to learn high-level semantic representations; MedMAE adapts this strategy to medical images; and SparK extends MIM to convolutional backbones by exploiting their multi-scale feature hierarchy.

Despite these advances, most MIM methods still rely on random or structure-agnostic masking, so the pretraining objective does not explicitly encode the semantic requirements of downstream

tasks. Non-selective masking can direct reconstruction capacity toward irrelevant visual regions and thereby weaken transfer. Recent region-aware methods attempt to restore semantic alignment, and the literature has developed along two principal directions:

The first direction is loss-driven adaptive masking. Methods such as hard patches mining (HPM) [Wang et al. (2023a)] and AnatoMask [Li et al. (2024)] estimate reconstruction difficulty across regions and assign higher masking probabilities to patches with larger losses, with the aim of prioritizing informative structures. Reconstruction difficulty, however, is not equivalent to task relevance: in brain magnetic resonance imaging (MRI), visually complex structures such as the spinal cord or eyeballs may receive excessive emphasis even when they are unrelated to downstream objectives such as tumor detection.

The second direction uses external perceptual modules to guide masking. FocusMAE [Basu et al. (2024)], for example, relies on pretrained Region Proposal Networks (RPNs) to identify high-information regions for selective masking. Although this strategy can be effective in specific settings, the RPN requires supervised training on domain-specific annotations, which limits cross-task portability.

These limitations motivate a masking mechanism that is flexible, semantically aligned, and usable without dense supervision. Natural language provides such a control signal: it is interpretable, compact, and able to specify task-relevant visual concepts without requiring per-image annotations.

2.2 VISION-LANGUAGE PRE-TRAINED MODELS

Recent vision-language alignment models make it increasingly feasible to identify task-relevant visual regions under zero-shot and open-vocabulary conditions [Radford et al. (2021)]. The Contrastive Language-Image Pretraining (CLIP) family learns a shared embedding space for images and texts through large-scale image-text pretraining and has become a standard foundation for zero-shot and open-set visual recognition. In medical imaging, BiomedCLIP [Zhang et al. (2023)] adapts cross-modal alignment to biomedical data and supports robust performance across downstream applications. Vision-language pretrained models have been widely used for supervised or downstream-oriented tasks such as object detection and semantic segmentation [Zhong et al. (2022); Aleem et al. (2024); Koleilat et al. (2024)]. Their capacity for open-vocabulary localization, however, has been used less directly as a mechanism for controlling the MIM pretraining process itself.

Together, these observations define the gap addressed by MWM. Region-aware MIM has improved semantic focus but remains constrained by reconstruction-error proxies, additional supervision, or limited flexibility. Vision-language models provide a complementary semantic control signal, yet prior work has rarely used natural language prompts to determine which image regions should be masked during pretraining. MWM closes this gap by incorporating vision-language semantic guidance into MIM, enabling efficient task-aware masking within an externally guided pretraining protocol.

3 METHODOLOGY

3.1 OVERALL FRAMEWORK

As illustrated in Figure 1, **Mask What Matters (MWM)** consists of two coupled stages: (1) text-guided region localization and (2) region-aware masked image modeling. The localization stage uses a frozen vision-language model to infer prompt-relevant regions from open-vocabulary text, and the MIM stage converts these regions into differentiated masking ratios that determine where reconstruction pressure is applied.

3.2 TEXT-GUIDED REGION LOCALIZATION

MWM localizes task-relevant regions through a sequential cross-modal pipeline. The process begins with prompt construction and uses BiomedCLIP to embed the image and prompt in a shared semantic space. The Multi-Modal Information Bottleneck (M2IB) then produces a saliency map, which is binarized by K-Means and spatially refined with the Segment Anything Model (SAM). The

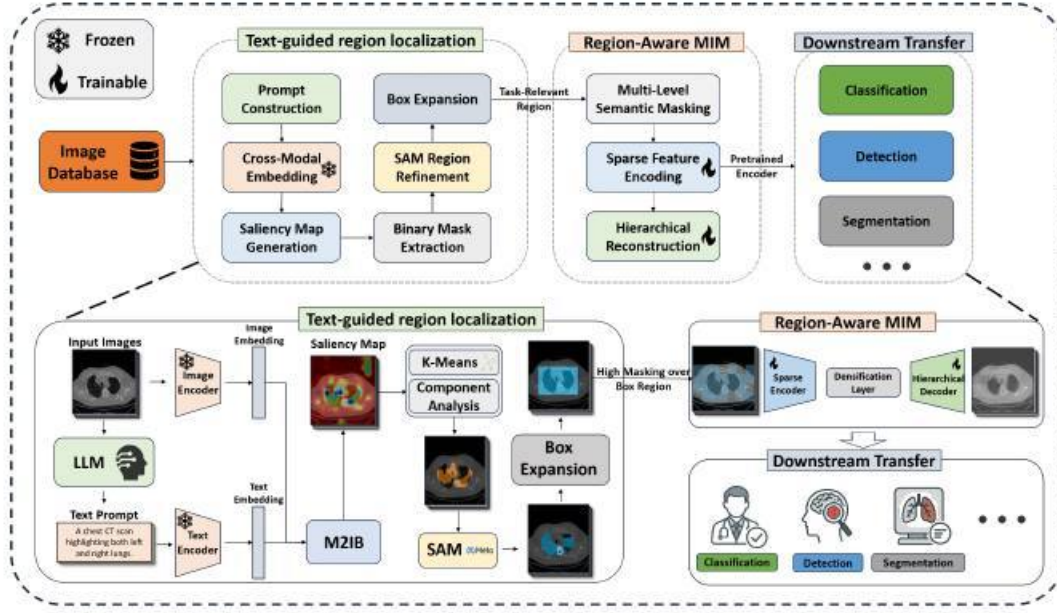


Figure 1: Overview of the MWM framework. The top panel summarizes the three-stage pipeline: (1) text-guided region localization, (2) region-aware masked image modeling, and (3) downstream transfer. The bottom panel shows how open-vocabulary prompts define localization cues that are converted into differentiated masking patterns.

refined mask is finally converted into a bounding box with a controllable expansion margin, yielding the spatial guide used by the masked modeling stage.

Prompt Generation. Textual prompt design determines the semantic specificity of region localization in medical images. To obtain descriptive guidance, large language models (LLMs) are queried with structured templates such as “Describe the typical visual characteristics of a *[category]* in a *[modality]* image”. Simple category phrases are also evaluated as alternative inputs, allowing the experiments to compare concise category-level prompts with richer descriptive prompts. These prompts provide semantic guidance at the concept level rather than per-image annotation.

Cross-Modal Embedding. After prompt construction, BiomedCLIP [Zhang et al. (2023)] encodes the medical image and corresponding prompt into a shared semantic space. For an image I and prompt T , the embeddings are computed as:

$$Z_{\text{img}} = \Phi_{\text{img}}(I), \quad Z_{\text{text}} = \Phi_{\text{text}}(T) \quad (1)$$

where Φ_{img} and Φ_{text} denote the vision and language encoders, respectively. The resulting embeddings define the cross-modal representation used by the subsequent saliency estimation step.

Saliency Map Generation. A cross-modal saliency estimation module based on the Multi-modal Information Bottleneck (M2IB) [Wang et al. (2023b)] localizes prompt-relevant visual regions by aligning image and text embeddings while suppressing modality-specific redundancy. Formally, the saliency map $\lambda_S \in [0, 1]^{H \times W}$ assigns an importance weight to each spatial location according to its semantic relevance to the prompt. M2IB estimates this map through the following objective:

$$\mathcal{L}_{\text{M2IB}} = \text{MI}(Z_{\text{img}}, Z_{\text{text}}) - \gamma \cdot \text{MI}(Z_{\text{img}}, I) \quad (2)$$

where Z_{img} and Z_{text} are the BiomedCLIP image and text embeddings, respectively, and γ controls the trade-off between preserving cross-modal relevance and suppressing task-irrelevant visual information.

Binary Mask Extraction. Given the saliency map, MWM applies unsupervised K-Means clustering [Lloyd (1982)] to obtain a preliminary foreground mask for the prompt-specified region. The map is partitioned into two pixel groups, and the group with higher saliency values is treated as foreground. Connected component analysis is then applied to the binary map; the largest foreground

regions are retained to suppress isolated false activations and concentrate the spatial guide on core semantic structures.

Region Refinement. SAM is used as a spatial refinement module to improve mask accuracy [Kirillov et al. (2023)]. For each selected connected region $c_i \in \mathcal{C}^*$, MWM computes the minimal enclosing bounding box and uses it as the visual prompt for SAM.

Given the original image I and the resulting bounding boxes, SAM predicts a refined segmentation mask:

$$M_{\text{SAM}} = \text{SAM}(I, \text{Box}(c_i)) \quad (3)$$

The resulting mask M_{SAM} provides a refined spatial approximation of the prompt-relevant region.

Box-Based Region Expansion. The SAM mask may still contain boundary irregularities that make the masking signal overly sensitive to local noise. To make region guidance more stable, the refined SAM mask is converted into a bounding box with a controllable expansion margin; this expanded box defines the final region of interest (ROI) for masked modeling.

3.3 REGION-AWARE MASKED IMAGE MODELING

Using the region of interest generated by the localization module in Section 3.2, MWM performs masked image modeling through three components: (1) multi-level semantic masking with region-dependent masking ratios, (2) sparse feature encoding over visible patches, and (3) hierarchical reconstruction from sparse multi-scale features.

Multi-Level Semantic Masking. MWM applies multi-level semantic masking to the regions identified by text-driven localization. Patches within prompt-highlighted regions, such as areas associated with tumors or organs, are masked at a higher ratio so that reconstruction emphasizes task-relevant semantic content. Background patches are masked lightly or left visible, reducing redundant computation while preserving contextual support. Figure 2 visualizes the resulting contrast between random and semantically guided masking patterns.

Sparse Feature Encoding. Following the SparK design [Tian et al. (2023)], visible patches are collected into a sparse image and passed to the encoder:

$$\begin{aligned} X_{\text{sparse}} &= \{x_i \mid m_i = 1\}, \\ F &= \Phi_{\text{sparse}}(X_{\text{sparse}}) \end{aligned} \quad (4)$$

where x_i denotes the i -th patch, $m_i \in \{0, 1\}$ indicates visibility, Φ_{sparse} is the sparse encoder, and $F = \{F_1, F_2, F_3, F_4\}$ denotes the resulting multi-scale feature set. Although the implementation instantiates MWM with convolutional backbones, the same masking interface can be applied to ViTs through token indexing.

Hierarchical Reconstruction. A lightweight UNet-style decoder reconstructs the full image through a top-down feature hierarchy. At each scale l , masked locations are densified with a learned embedding M_l :

$$F'_l(i) = \begin{cases} F_l(i), & \text{if patch } i \text{ is visible,} \\ M_l, & \text{if patch } i \text{ is masked.} \end{cases} \quad (5)$$

The densified feature map F'_l is projected by the scale-specific layer ϕ_l to produce D_l , which is then upsampled and fused with features from the next lower level:

$$D_{l-1} = B_{l-1}(D_l) + \phi_{l-1}(F'_{l-1}). \quad (6)$$

The reconstruction objective is the mean-squared error over masked patches:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{I}_{\text{mask}}|} \sum_{i \in \mathcal{I}_{\text{mask}}} \|\hat{x}_i - x_i\|^2, \quad (7)$$

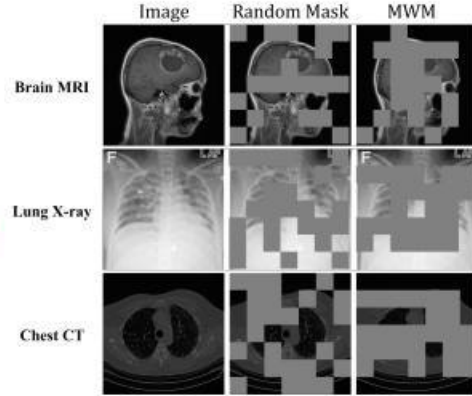


Figure 2: Comparison of masking strategies, with gray blocks denoting masked regions.

where $\mathcal{I}_{\text{mask}}$ denotes the set of masked patch indices, x_i is the original patch, and $\hat{x}_i$ is the reconstructed patch.

4 EXPERIMENTS

4.1 DATASETS

Self-Supervised Pretraining Datasets. Self-supervised pretraining uses three datasets that cover complementary medical imaging modalities and provide a basis for evaluating prompt-guided masking beyond a single image type. (1) *Brain MRI* contains approximately 17,000 images from Brain Tumor MRI [Cheng et al. (2015)], BRISC [Fateh et al. (2025)], and BraTS 2018 [Menze et al. (2014)], covering multiple tumor types and anatomical structures. (2) *Lung X-ray* includes around 39,000 chest radiographs from COVID-QU-Ex [Rahman et al. (2021)]; Chowdhury et al. (2020), spanning COVID-19, non-COVID pneumonia, and normal cases. (3) *Chest CT* comprises 12,000 slices from a lung disease dataset [Konya (2020)], including segmentation masks of fibrotic lungs from 107 patients.

Downstream Fine-Tuning Datasets. Downstream evaluation uses three datasets to assess classification transfer across distinct anatomical regions and imaging protocols. (1) *Brain Tumor MRI* [Felltrin (2023)] contains 4,479 MRI images across 44 tumor types, including astrocytoma, glioblastoma, meningioma, and ependymoma. (2) *Pediatric Lung X-ray* [Kermary et al. (2018)] consists of 5,863 pediatric chest X-rays categorized as normal or pneumonia. (3) *Chest CT Cancer* [Hany (2020)] includes CT images from four classes: adenocarcinoma, large cell carcinoma, squamous cell carcinoma, and normal tissue.

4.2 EXPERIMENTAL SETUP

All self-supervised methods are evaluated with backbones of comparable scale: convolutional methods use ResNet-50 [He et al. (2016)], and transformer-based methods use ViT-S [Dosovitskiy et al. (2020)]. Images are resized to 224×224 , and all models are initialized from ImageNet-pretrained weights. Pretraining uses an initial learning rate of 2×10^{-4} ; downstream fine-tuning preserves the encoder and input resolution. Experiments are conducted on NVIDIA Tesla A100 GPUs with 80 GB VRAM using Python 3.9, Ubuntu 22.04.3, and PyTorch 1.10.0. Unless otherwise specified, non-target variables such as prompt type and SAM usage are fixed to their empirically selected settings so that each ablation isolates the factor under evaluation. These comparisons therefore evaluate the proposed masking strategy under a shared downstream protocol, while MWM additionally uses external pretrained localization components that are not provided to the SSL baselines.

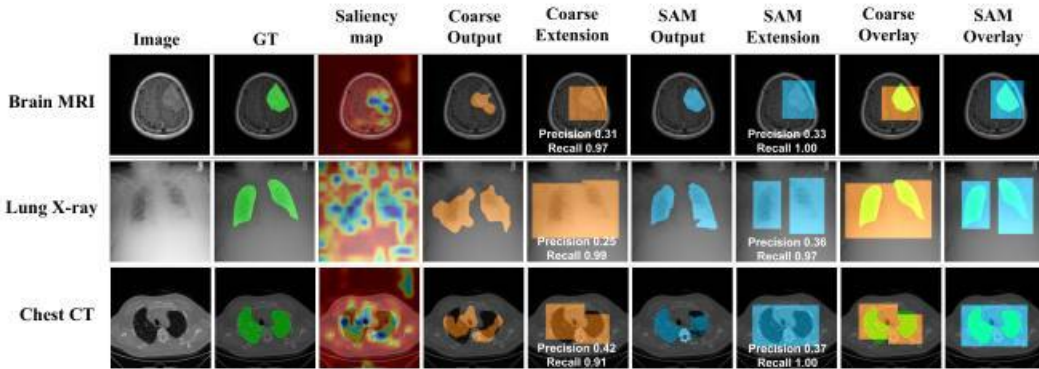


Figure 3: Representative text-guided localization results across the evaluated medical imaging modalities.

Method	Brain MRI		Chest CT		Lung X-ray	
	Prec.	Rec.	Prec.	Rec.	Prec.	Rec.
<i>Prompt Type</i>						
Phrase	0.22	0.86	0.21	0.99	0.42	0.82
Descrip. Sent.	0.30	0.94	0.37	0.88	0.44	0.66
<i>Impact of SAM</i>						
Coarse Ext.	0.22	0.95	0.36	0.87	0.37	0.84
+ SAM Ext.	0.30	0.94	0.37	0.88	0.42	0.82

Table 1: Text-guided localization performance measured by Occlusion Precision and Recall across prompt types and refinement strategies. *Coarse Ext.* denotes bounding-box extension of the coarse region; + *SAM Ext.* denotes additional SAM-based refinement.

4.3 EVALUATION OF TEXT-GUIDED LOCALIZATION

We first evaluate whether the text-guided localization module identifies task-relevant regions across brain MRI, lung X-ray, and chest CT, and then isolate the effects of prompt type and SAM refinement. Predicted bounding boxes are compared with expert-annotated segmentation masks using occlusion precision ($|R_p \cap R_{gt}|/|R_p|$) and occlusion recall ($|R_p \cap R_{gt}|/|R_{gt}|$), where R_p denotes the predicted region and R_{gt} denotes the ground-truth annotation. These metrics evaluate whether the localization stage supplies useful spatial guidance for masking rather than serving as a standalone clinical segmentation system.

Table 1 shows that text-guided localization achieves high occlusion recall across all three modalities while maintaining usable, though not exhaustive, precision. The result indicates that prompt-derived semantic cues are sufficient to recover substantial portions of task-relevant regions despite anatomical and modality-specific variation; qualitative examples are shown in Figure 3.

Effect of Prompt Type on Region Localization. To assess how language granularity affects localization, we compare descriptive sentence prompts with short category phrases. Table 1 indicates that descriptive sentences improve precision on Brain MRI and Chest CT, where richer semantic specification helps identify localized structures. For lung radiographs, concise phrases yield higher recall (0.82 vs. 0.66), suggesting that broad anatomical targets can be captured effectively by simpler textual descriptions.

Effect of SAM Refinement on Region Localization. We next measure the contribution of spatial refinement by applying SAM to the coarse masks. Table 1 shows that SAM improves precision while preserving high recall; on Brain MRI, for example, precision increases from 0.22 to 0.30 while recall remains above 0.94. These results support the use of SAM as a refinement step for converting coarse cross-modal saliency into more reliable masking regions, while also indicating that the localization signal remains imperfect and should be interpreted as guidance for pretraining rather than definitive delineation.

4.4 COMPARISON WITH PREVIOUS SSL METHODS

We next evaluate the transferability of MWM representations by comparing them with representative SSL methods from reconstruction-based and contrastive-learning paradigms. All methods are pre-trained and evaluated under consistent datasets and training settings, so performance differences are measured under a shared evaluation protocol. Because MWM uses frozen external vision-language and segmentation models to construct its masks, the comparison primarily tests the benefit of injecting semantic localization into MIM rather than isolating masking alone from all external priors.

Image Classification. Classification transfer is evaluated on three downstream medical imaging datasets under two complementary protocols: (1) *full fine-tuning*, where all model parameters are updated, and (2) *linear probing*, where the encoder remains frozen and only a linear classifier is trained.

- **Full fine-tuning:** Table 2 reports classification accuracy for MWM and representative self-supervised methods, including MAE [He et al. (2022)], SparK [Tian et al. (2023)], AnatoMask [Li

et al. (2024), SimCLR (Chen et al. (2020a)), MoCoV2 (Chen et al. (2020b)), and BYOL (Grill et al. (2020)). MWM achieves the highest accuracy on all three datasets, indicating that text-guided masking transfers consistently across the evaluated modalities. The largest margin appears on Chest CT, where MWM reaches 97.5% accuracy, exceeding SparK by +3.1 points and AnatoMask by +1.7 points. This gain is consistent with the intended role of MWM: the masking policy concentrates reconstruction on semantically critical and spatially sparse regions, such as lung lobes in CT scans, while reducing redundant background masking.

- **Linear probing:** Linear probing further probes representation quality by freezing the encoder. MWM is compared with two recent MIM baselines, SparK and AnatoMask. As shown in Table 3, MWM obtains the best frozen-encoder accuracy on every dataset, supporting the conclusion that text-driven pretraining improves the learned feature space rather than only benefiting end-to-end fine-tuning.

Object Detection and Instance Segmentation. To test whether the learned representations extend beyond classification, we evaluate brain tumor detection and instance segmentation on the BR35H dataset (Hamada (2025)). Table 4 shows that MWM achieves the highest AP^{box} (46.9%) and the second-highest AP^{mask} (45.9%), exceeding SparK by +1.3 and +1.1 points, respectively. The object-level and pixel-level results indicate that the benefits of text-guided masking extend to localization-oriented downstream tasks as well as classification, although AnatoMask remains marginally higher on AP^{mask} .

Overall, the comparative experiments show that text-driven masking produces more informative and transferable representations under the evaluated protocols. The consistent improvements across classification, detection, and segmentation support the central claim that semantic control over the masking process strengthens medical visual representation learning when prompt-level guidance and pretrained localization models are available.

4.5 ABLATION STUDY

Ablation experiments isolate how prompt design, SAM-based region refinement, and masking ratio contribute to MWM classification performance. These analyses connect the localization mechanism to downstream transfer and clarify which components are responsible for the observed gains.

4.5.1 EFFECT OF PROMPT DESIGN AND REGION REFINEMENT

As shown in Table 5, descriptive sentence prompts outperform short phrases on brain MRI and chest CT, whereas lung X-ray favors the phrase prompt. This pattern is consistent with the localization analysis in Section 4.3: prompt forms that better identify the relevant region tend to yield stronger downstream accuracy. Both prompt types, despite their different granularity, outperform the no-prompt random masking baseline on every dataset. This result demonstrates that text-guided masking improves the semantic alignment between pretraining and downstream objectives under the evaluated prompt choices. The SAM ablation further shows that incorporating spatial refinement during pretraining improves classification accuracy across modalities, validating its role in strengthening the region guidance used by MWM.

Method	Type	Brain MRI	Chest CT	Lung X-ray
MoCoV2	CL	95.3	91.7	93.1
BYOL	CL	91.3	93.7	93.9
SimCLR	CL	94.5	93.1	94.1
MAE	MIM	95.4	91.7	94.9
AnatoMask	MIM	96.5	95.8	95.2
SparK	MIM	96.2	94.4	94.7
MWM(Ours)	MIM	96.8	97.5	96.0

Table 2: Downstream fine-tuning classification accuracy (%) across modalities. MIM denotes Masked Image Modeling, and CL denotes Contrastive Learning.

Method	Brain MRI	Chest CT	Lung X-ray
AnatoMask	70.2	63.9	85.3
SparK	69.8	63.5	81.6
MWM(Ours)	71.3	67.0	88.5

Table 3: Linear probing accuracy (%) on downstream datasets with the encoder frozen.

Method	Detection		Segmentation	
	AP^{box}	AP_{75}^{box}	AP^{mask}	AP_{75}^{mask}
AnatoMask	45.2	53.1	46.2	54.1
SparK	45.6	53.2	44.8	52.0
MWM (Ours)	46.9	53.5	45.9	54.1

Table 4: Detection and instance segmentation results (%) on BR35H, reported as AP^{box} , AP_{75}^{box} , AP^{mask} , and AP_{75}^{mask} .

4.5.2 EFFECT OF MASKING RATIO

Figure 4 shows that MWM reaches 96.8% classification accuracy on Chest CT with a 40% masking ratio, well below the approximately 70% ratio commonly used in MIM. Even with this reduced masking level, MWM exceeds SparK’s best 70% masking result by +2.4 percentage points. To ensure that this comparison is not explained by an unfavorable SparK configuration, SparK is evaluated at 40%, 50%, 60%, and 70% masking ratios. The best SparK result occurs at 70%, indicating that the reported MWM advantage is not an artifact of selecting a weak baseline ratio.

These findings refine the usual assumption that higher masking ratios are intrinsically preferable for MIM. In the evaluated Chest CT setting, semantic control over which regions are masked matters more than simply increasing the amount of masked content: MWM preserves useful context, reduces redundant masking, and concentrates reconstruction on task-relevant structures.

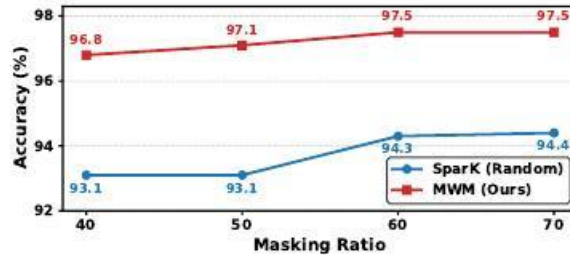


Figure 4: Chest CT classification accuracy under different masking strategies and masking ratios.

5 CONCLUSION AND LIMITATIONS

This paper presents **Mask What Matters (MWM)**, an externally guided self-supervised pretraining framework that couples text-guided semantic localization with region-aware masking. By using natural language prompts to identify task-relevant regions and applying differentiated masking to those regions and the background, MWM produces semantically aligned representations without relying on per-image annotations.

The experiments substantiate three conclusions within the evaluated medical imaging settings: (1) natural language prompts provide useful semantic cues for region localization, establishing the feasibility of text-driven masking; (2) MWM outperforms existing methods such as SparK and AnatoMask in classification across three modalities and improves most detection and instance segmentation metrics; (3) MWM maintains strong performance at lower masking ratios, including 40%, showing that semantic masking control can be more effective than uniformly increasing the masking ratio.

Limitations. The framework also has clear boundaries. First, although text prompts provide flexible guidance, the robustness of MWM to variations in prompt style, vocabulary, or noise has not been systematically evaluated. Second, the use of external textual descriptions and frozen pretrained

Group	Setting	Brain MRI	Chest CT	Lung X-ray
Prompt Type	No Prompt	96.2	94.4	94.7
	Phrase	96.4	96.5	96.0
	Sentence	96.8	97.5	95.2
Impact of SAM	w/o SAM	96.4	96.8	95.8
	w/ SAM	96.8	97.5	96.0

Table 5: Downstream classification accuracy (%) under alternative prompt types and region refinement strategies across three imaging modalities. *No Prompt* denotes random masking without text-guided localization. *w/* and *w/o* denote with and without SAM, respectively.

models introduces dependencies that may reduce reliability when prompts are unavailable, inconsistent, poorly matched to the target task, or when localization quality degrades. Third, the current experiments report single-run performance without variance estimates or statistical significance testing, so the quantitative improvements should be interpreted as empirical trends within the reported protocols rather than distributional guarantees.

Looking forward, text-driven region-aware masking offers a principled direction for semantically grounded self-supervised learning. Extending MWM beyond the evaluated medical imaging modalities, and evaluating it under prompt sets and external-information budgets that more closely match competing baselines, remains an important opportunity for future work on vision-language-guided pretraining.

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A USE OF LARGE LANGUAGE MODELS (LLMs)

LLMs were used *only* for language editing and presentation polishing of author-written text, including grammar, phrasing, and minor stylistic clarification. LLMs were *not* used for idea generation, experimental design, data annotation, code implementation, or analysis/interpretation of results. All technical content, algorithms, proofs, figures, tables, metrics, and conclusions were authored and verified by the authors. No confidential or identifying data were provided to LLMs beyond anonymized manuscript text.